68. *On the Direction of Magnetisation of Basalt in Japan, Tyôsen and Manchuria.*

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Early in April, 1926, a specimen of basalt from Genbudô, Tazima, a celebrated basalt cave, was collected for the purpose of examining its magnetic properties. Its orientation was carefully measured in its natural position before it was removed. When this block was tested by bringing near to a freely suspended magnetic needle, its magnetic north pole was found to be directed to the south and above the horizontal direction. This is nearly opposite to the present earth's magnetic field at the locality. In May of the same year, four specimens of basalt were collected from Yakuno, Tanba, with the similar care. When tested, their magnetic axes were found to have an easterly declination of some 20° and a downward inclination of some 50°.

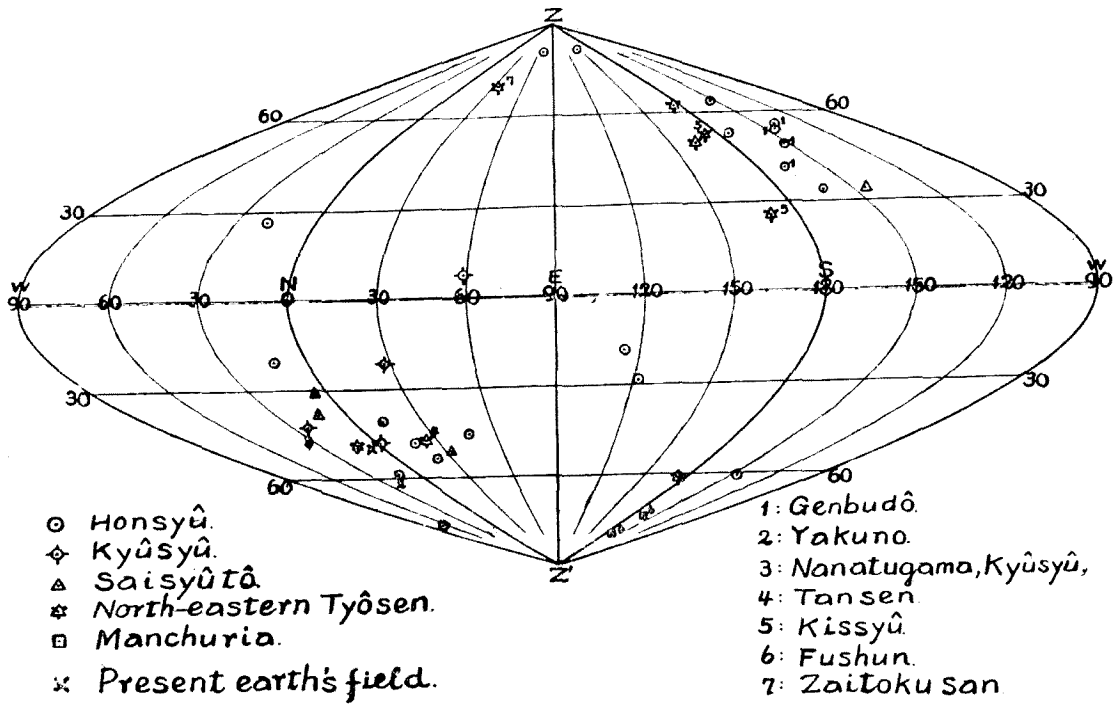
Since the time of Melloni[1] it is believed that lava gets its magnetism in cooling in the direction of the earth's magnetic field. This was also proved experimentally by Prof. Nakamura[2]. The above mentioned places are not much distant from each other and have nearly the same magnetic field. These basalts are described as the lavas of probably Quarternary eruptions[3].

Since that time 139 specimens of basalt were collected from 36 places in Honsyû, Kyûsyû, Tyôsen and Manchuria, of which 38 specimens were already examined more accurately. The method depended upon Gauss's principle of analysing the earth's magnetism, which was also used by Prof. Nakamura. The specimen was enclosed and fixed in a spherical surface in such a way that its orientation could be read from outside. Distribution of the normal component of magnetic force on the surface of the sphere due to the enclosed basalt was determined by means of a magnetometer and the direction of magnetic axis and the intensity of magnetisation were determined by the method of harmonic analysis.

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1) Förstemann, F.C., Poggendorf Ann., **106** (1859), 106–136.
2) Nakamura, S., Tôkyô Sûgaku-Buturigakkwai Kizi, [2] **6** (1912), 268–273.
3) 巨智部忠承, 豊岡圖幅地質說明書, 明治二十八年.
,, 生野圖幅地質說明書, 明治二十九年.
鈴木 醇, 地理學評論, 第一卷, 356–365, 大正十四年.

Direction of North Magnetic Pole of Basalt. (M. Matuyama)



To see the degree of reliability, 4 specimens from Genbudô were examined. As the result their magnetic axes were found to have a mean westerly declination of 150° and a mean upward inclination of 49°, the largest deviations being 12° in the former and 14° in the latter.

Care was also paid to the tilting of the crust after the basalt sheet was laid out. At Fushun, Manchuria, the specimens were taken from a dolerite sheet which was conformably under and overlaid by a shale strata dipping 25° to the north with a nearly eastwesterly strike[1]. Hence due corrections were made for the measured direction of magnetic axes of these specimens.

The measured directions of magnetic north pole are plotted in the figure. The earth's magnetic field in the related area has a westerly declination varying from 4°.5 to 6°.5 and a downward inclination from 40°.5 to 60°. Since much larger variations of angles are concerned here, the mean direction was considered to prevail over this area.

From the figure, a peculiar arrangement can easily be noticed. There is a group of specimens, including that from Yakuno, whose directions of magnetisation falls around the present earth's field. A number of other specimens, including that from Genbudô, forms another group almost exactly antipodal to the former. The remaining

1) Makiyama, J., Japanese Jour. Astr. Geoph., **2** (1924), 9-11.

specimens which belong to neither of these groups are not so numerous so far as it was examined.

The age of eruption of the collected basalts are not always clearly known. Among the specimens of the first group, those from the north-western Kyûsyû are described to be of post-Tertiary[1] and that from Tansen, Tyôsen, is believed to be of Quarternary period. None of them is known to be older than the beginning of the Quarternary; they might be younger. In the second group, the basalt of Genbudô is considered to be also of Quarternary eruption. One of the specimens from Kissyû, Tyôsen, is reported to be of Pleistcene[2] while the other is definitely older, since it is surrounded and a part covered by the first. Thus we may consider that in the earlier part of the Quarternary period the earth's magnetic field in the area under consideration was probably in the state represented by the second group, which gradually changed to the state represented by the first group.

There are three examined specimens of basalt known as of Tertiary eruption. Two from Fushun, Manchuria, had mean westerly declination of 168° and downward inclination of 77°. They are described to be of Miocene period. A specimen from Zaitokusan, Tyôsen, is reported to form the base of the Sitihôzan group unconformably covering the probably Oligocene group[3], and may be of Miocene. When examined this specimen had easterly declination of 35° and upward inclination of 70°. Thus there is also a roughly antipodal relation.

According to Mercanton[4] the earth's magnetic field was probably in a greatly different or nearly opposite state in Permo-carboniferous and Tertiary ages compared to the present. From my results it seems as if the earth's magnetic field in the present area has changed even to opposite direction in comparatively shorter duration, in Miocene and also in Quarternary periods.

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1) 大塚傳一, 佐賀圖幅地質說明書, 明治三十四年.
大築洋之助, 平戸圖幅地質說明書, 大正六年.
2) Yamanari, F., Geological Atlas of Chosen, No. 3, 1925.
3) Tateiwa, I., Geological Atlas of Chosen, No. 4, 1925.
4) Mercanton, P.L., Comptes Rendus, **182** (1926), 859–861 and 1231–1232.